

scripts/autonomous-qa/lib-jackett.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)

Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.B ("Up" is not healthy), §6.H (no secret echo), §6.J (fail loudly), §6.R (no-hardcoding), §6.U (no sudo) **Classification:** project-specific

Overview

Sourceable library that brings up / tears down the **Jackett (Torznab) + FlareSolverr (Cloudflare-solver) sidecar** so lava-api-go can serve Cloudflare-protected tracker searches (notably IPTorrents) in the Phase-2 autonomous-QA matrix.

It is a thin autonomous-QA wrapper around the SAME mechanism the project's canonical validator (`scripts/validate-jackett-sidecar.sh`) uses: rootless podman (preferred) or docker — NO sudo — driving `tools/lava-containers/docker-compose.jackett.yml` with `--profile jackett` `--profile cloudflare`. Jackett/FlareSolverr ports bind to host **loopback only**; lava-api-go (host-net) reaches Jackett at `http://127.0.0.1:9117` and holds the `api_key` server-side. The Android app NEVER talks to Jackett (§6.H).

Raw `<runtime> compose` is correct here (the fragment is `bridge + loopback, no network_mode: host`); `./start.sh + BUILDDAH_FORMAT=docker` are only needed for the host-net api-go profile whose image is locally built — the Jackett / FlareSolverr images are pulled with HEALTHCHECKs baked in.

Functions

Function	Purpose
<code>jackett_up</code>	Bring the sidecar up and BLOCK until the real Torznab surface answers, or fail loudly on timeout

Function	Purpose
jackett_down	Tear the sidecar down (both profiles, idempotent)

Usage

```
source scripts/autonomous-qa/lib-jackett.sh
jackett_up      # bring up jackett + flaresolVERR, block until
                ready
# ... run Phase-2 IPTorrents searches against lava-api-go ...
jackett_down    # tear it all down
```

Env knobs (all optional)

Variable	Default	Meaning
LAVA_QA_CONTAINER_RUNTIME	auto-detect (podman → docker)	Force podman or docker
LAVA_QA_JACKETT_CLOUDFLARE	1	0 = jackett only (skip the RAM-heavy FlareSolvErr)
LAVA_QA_JACKETT_TIMEOUT	120	Readiness timeout, seconds

Every host/port/api_key/image/indexer comes from .env (placeholders in .env.example) per §6.R — LAVA_API_JACKETT_APIKEY, LAVA_API_JACKETT_DEFAULT_INDEXER, LAVA_JACKETT_HOST_PORT, LAVA_FLARESOLVERR_HOST_PORT, LAVA_JACKETT_BIND_HOST, ...

Readiness contract (anti-bluff §6.B / §6.J)

jackett_up does NOT assert container State==running; it polls the REAL user-visible surfaces until they answer, and FAILS LOUDLY (with logs) on timeout:

- **Jackett** — GET .../results/torznab/api?t=caps&apikey=<key> must return HTTP 200 with parseable Torznab XML; an auth-error body (incorrect api key) is rejected explicitly. The api_key is REDACTED in every log line (§6.H). Without a valid key in .env the function refuses to run — an unauth probe would prove nothing.
- **FlareSolvErr** (cloudflare profile only) — POST /v1 {"cmd":"sessions.list"} → 200 {"status":"ok"} (Chromium warm-

up is slow; this is the de-facto liveness probe — FlareSolvrr has no GET /health).

Return: 0 = sidecar READY (probed); non-zero = failed (container logs dumped, sidecar torn down).

Known unconfirmed value

The 120 s readiness default is **UNCONFIRMED** against the target gate host — the compose fragment itself notes the .NET/Chromium cold-start seconds are placeholders. Measure real cold-boot on the Linux x86_64 gate host and tune LAVA_QA_JACKETT_TIMEOUT once observed; do not assume.

Companion files

- scripts/validate-jackett-sidecar.sh (+ its doc) — the canonical validator
- tools/lava-containers/docker-compose.jackett.yml — the compose fragment
- docs/qa/jackett-sidecar-deployment.md — sidecar deployment guide